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Modeling and analysis of high-frequency alternating-current heating for lithium-ion batteries under low-temperature operations

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HIGHLIGHTS

- A high-frequency alternating-current heating strategy is proposed for cold batteries.
- A thermoelectric model considering heat generation of charge transport is developed.
- The heating speed and efficiency are improved by increasing switching frequency.
- The design of the inductance and switching frequency is provided.
- The heating speed at different frequencies and currents is predicted.

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ABSTRACT

In cold climates, preheating is necessary to improve the output power and available capacity of low-temperature lithium-ion batteries. Many internal Alternating Current (AC) heating approaches are available to heat low-temperature batteries with the advantages of fast heating speed, high efficiency, and good uniformity. However, due to the lower AC heating frequencies, the existing AC heating devices have the prominent features of bulky size and high cost, leading to difficult onboard implementation in electric vehicles. Therefore, this paper proposes a high-frequency AC heater based on buck-boost conversion and develops the high-frequency thermoelectric model based on the heat generation of the ohmic resistance and lithium ion transport. The experimental results verify the validities of the proposed high-frequency AC heating topology and thermoelectric model at different AC-heating frequencies and Root-Mean-Square (RMS) currents. This study shows increasing the AC-heating frequency at the same RMS current can dramatically improve the heating speed and efficiency due to the increased heat generation of the ohmic resistance and lithium ion transport, which does not cause further damage to batteries.

1. Introduction

In recent years, in order to save energy and protect the environment, the world is striving to develop Electric Vehicles (EVs) [1]. As the power source of EVs, the performances of lithium-ion batteries have significant effect on the driving range of EVs [2]. Particularly, in cold climate, the internal resistance of lithium-ion batteries dramatically increases [3], causing significant loss of the output power and available capacity of battery packs [4], which leads to a tremendous decrease of the EV driving range [5]. For example, the American Automotive Association (AAA) Automotive Research Center showed that the driving range of EV decreased by 60% in extreme temperatures [6]. In addition, the ageing rates of lithium-ion batteries gradually increase with the decrease of temperature [7]. Therefore, Herreyre et al. [8] invented new lithium-ion electrolytes to obtain good charging and discharging performances below $-30\,^{\circ}C$ in different $LiCoO_2||$graphite battery cells. In fact, the most effective solution to improve the performances of low-temperature batteries is not only from the modification of new lithium-ion electrolytes [9], but also to use preheating techniques to heat cold lithium-ion batteries to room temperature [10]. Wang et al. [11] indicated that the preheating could reduce the operating cost of Plug-in Hybrid Electric

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Nomenclature

B	Battery cell
V_B	Battery voltage (V)
C	Battery capacity (Ah)
m	Battery mass (g)
S	Battery surface area (m^2)
c	Specific heat capacity ($J.g^{-1}.k^{-1}$)
h	Equivalent heat transfer coefficient ($W.m^{-2}.k^{-1}$)
T_0	Ambient temperature (°C)
T_B	Battery temperature (°C)
L	Inductor (H)
L_B	Battery inductance (H)
R_B	Battery internal resistance (Ω)
R_L	Inductor resistance (Ω)
$R_{DS(on)}$	Static drain-source on resistance (Ω)
R_{eq}	Equivalent resistance (Ω)
Q_1-Q_4	MOSFETs
V_{oc}	Open circuit voltage (V)
i_L	Inductor current (A)
$I_{L(RMS)}$	RMS inductor current (A)
T_{SW}	Switching period (s)
f_{SW}	Switching frequency (Hz)
i_B	Battery heating current (A)
$I_{B(RMS)}$	RMS battery heating current (A)
$I_{B(max)}$	Maximum discharging current (A)
t	Time (s)
R_1	Electrochemical polarization resistance (Ω)
C_1	Electric double-layer capacitance (F)
η_h	Heating efficiency (%)

Abbreviations

AC	Alternating Current
DC	Direct Current
AAA	American Automotive Association
EV	Electric Vehicle
PHEV	Plug-in Hybrid Electric Vehicle
RMS	Root Mean Square
OCV	Open Circuit Voltage
KCL	Kirchhoff's Current Law
KVL	Kirchhoff's Voltage Law
LC	Inductor Capacitor
PWM	Pulse Width Modulation
RC	Resistor Capacitor

Vehicles (PHEVs) by up to 22.3%.

Many preheating approaches are now available to heat cold batteries and improve the battery working efficiency, which mainly consist of the external heating and internal heating approaches [12]. The external heating solution usually employs electric resistance wires to externally heat batteries by the medium of air or liquid [13], leading to a low heat conduction efficiency, long heating time, and bad heating consistency. Particularly, it needs an external power source, e.g., the charging point, which is difficult to be implemented in EVs [13]. In contrast, the internal Alternating Current (AC) heating approach uses an AC current to charge and discharge batteries, so that the heat generation of the ohmic resistance and the electrochemical reaction can internally warm batteries [14]. Without the need of the heat conduction, this approach significantly reduces the heat dissipation to the environment [14]. Therefore, the internal heating approach is more promising to provide a faster and more efficient heating for cold batteries with a better temperature uniformity than the external heating method [13].

Pesaran et al. [15] compared the external and internal heating approaches and concluded that the internal AC heating solutions had a faster heating speed and more consistent temperature-rise than the external methods. Moreover, the AC heating had less effect on battery health. Ji et al. [16] proposed three heating strategies, i.e., self-internal heating, convective heating and mutual pulse heating, and discussed their advantages and disadvantages in terms of capacity loss, heating time, system durability, and cost. This study indicated that a high-frequency AC current with a large amplitude is recommended to offer both high heating speed and long battery cycle life. Yang et al. [17] compared the external and internal heating solutions in terms of the heating speed and safety. They indicated that in order to prevent local over-heating of batteries, the heating power of the external heating approach is limited, leading to a very slow heating speed [17]. Therefore, recently, the internal AC heating is extensively studied because of the fast heating speed and high efficiency.

Zhang et al. [18] developed a temperature-rise model for the lower-frequency (lower than 10 Hz) AC heating and indicated that the heating speed can be improved with a larger amplitude or a lower heating frequency. However, due to the lower AC-heating frequency, this heating approach cannot be directly applied to EVs because of the bulky size and high cost. Jiang et al. [19] built a thermoelectric coupling model to simulate the electric and thermal behaviors of batteries in extreme environment. Mohan et al. [20] proposed a heating technique based on predictive control and optimized the amplitude of the AC current to maximize the heating efficiency. Ruan et al. [21] proposed an electro-thermal coupled model to optimize the heating frequency to achieve a good tradeoff between the fast heating speed and little negative effect on battery health. However, the electro-thermal coupled model did not include the heat generation of the high-frequency charge transport. Zhu et al. [22] studied the effect of the AC current frequencies, amplitudes, and voltage limitations on the internal heating performance. Due to the large impedance, the low AC heating frequency was a good choice to heat cold batteries. Hundreds AC heating cycles were conducted to validate that the AC pulse heating did not aggravate battery degradation. Zhang et al. [23] proposed a control strategy to internally heat the battery during regenerative braking and rest periods of driving. Simulation results show that the driving range of a vehicle was improved by 49% at −40 °C. Jiang et al. [24] proposed a self-heating approach superimposing the discharge current in the alternating current to prevent lithium-ion deposition, which was verified by 600 repeated heating tests. Guo et al. [25] proposed an echelon preheating strategy, which could calculate the optimal heating current based on the electro-thermal coupled model, to achieve a high temperature-rise rate for cold batteries. Experiment results showed that this AC heating strategy did not cause apparent damage on batteries. It is worth mentioning that Wang et al. [26] reported an all-climate lithium-ion battery, which only took 20 s to be self heated from −20 °C to 0 °C, consuming only 3.8% of battery capacity. Then, Zhang et al. [27] improved the self-heating lithium-ion battery, which can be self heated from −20 °C to 0 °C within 12.5 s. However, the reliability and the commercial application of this new battery structure need to be further tested.

In conclusion, the existing AC heating approaches focus on the low-frequency (generally lower than 3 kHz) AC heating, which are inconvenient to be utilized in electric vehicles (EVs) because of the bulky size, high cost, slow heating speed, harm to batteries, and low efficiency. Fortunately, Stuart and Hande [28] designed a high-frequency AC battery heater. However, the heater was powered by the on-board generator, which could not be used in EVs. In addition, Shang et al. [29] proposed a high-frequency sine-wave heater based on resonant Inductor-Capacitor (LC) converters to heat the automotive batteries at low-temperatures without the need of external devices. However, in order to achieve the sinusoidal heating current, the AC frequency and current amplitude could not be adjusted online to adapt to different heating requirements. As a result, an automotive onboard AC heater based on buck-boost conversion was proposed in Ref. [30] and the heating experiments were conducted at different heating frequencies. Contrary to the conclusions in Ref. [18], this study showed that the heating process could be speeded up by increasing the AC-heating frequency at the same current amplitude. This implied that the heat generation of the AC heating was highly sensitive to the AC-heating frequency. However, this study failed to build the model of the high-frequency AC heating and discuss how to optimize the AC current and frequency.

In fact, when a battery is AC heated, lithium ions are rapidly transferred from cathode to anode and vice versa, which continuously generate a large amount of heat inside the battery [31]. Apparently, the higher the AC-heating frequency, the faster the lithium ion transport, and the more the heat generation [32]. Therefore, increasing the AC-heating frequency can significantly increase the heat generation of the lithium ion transport.

In summary, it is optimal to heat low-temperature batteries at higher AC-heating frequencies. Firstly, a higher AC-heating frequency can significantly increase the heat generations of both the ohmic resistance and the charge transport [33], leading to a relatively higher heating speed and efficiency. Secondly, the higher the AC-heating frequency, the more difficult the lithium separation, and the less the harm to batteries [34]. Finally, the size and cost of the AC heater can be dramatically reduced at higher AC-heating frequencies [35], leading to an easy onboard implementation in EVs.

Therefore, this paper develops a high-frequency AC heating strategy and thermoelectric model, which is also a theoretical extension of the automotive onboard AC heater based on buck-boost conversion [30]. Specifically, this paper has two original contributions. Firstly, the analytical expression of the AC heating current of the buck-boost-based heater is derived to guide the design of the inductance and switching frequency. Secondly, a high-frequency thermoelectric model considering the heat generation of the charge transport is developed to accurately predict the heating speed at different AC-heating frequencies and RMS currents. Finally, it is theoretically and experimentally verified that increasing the AC-heating frequency can significantly improve the heating speed and efficiency without causing more lithium-ion deposition, leading to an easy and cheap implementation of low-temperature battery heating in EVs.

2. Methodology

In order to verify the validity of the proposed thermoelectric model at different switching frequencies and Root-Mean-Square (RMS) currents, a high-frequency AC heater is necessary to generate an adjustable-frequency or -amplitude AC current to internally heat batteries. It is worth mentioning that the automotive onboard AC heater based on buck-boost conversion [30] can be easily used to heat batteries at different AC-heating frequencies and RMS currents, because it has the favorable performances as follows:

(1) With the same inductance, the heating current can be easily adjusted online by controlling the switching frequency.
(2) At the same switching frequency, the heating current can be also regulated by changing the inductance.

2.1. Modeling of the AC heater based on buck-boost conversion

The proposed buck-boost-based AC heater has two structures, i.e., the half-bridge [30] and full-bridge structures. As shown in Fig. 1 (a), the half-bridge structure with two switches can be applied to an even (two)-cell battery pack, which can realize the alternate heating for the two-part cells. Nevertheless, for the odd (three)-cell battery string, the full-bridge structure with four switches can be used to realize the full-time heating for all the cells, as shown in Fig. 1 (b).

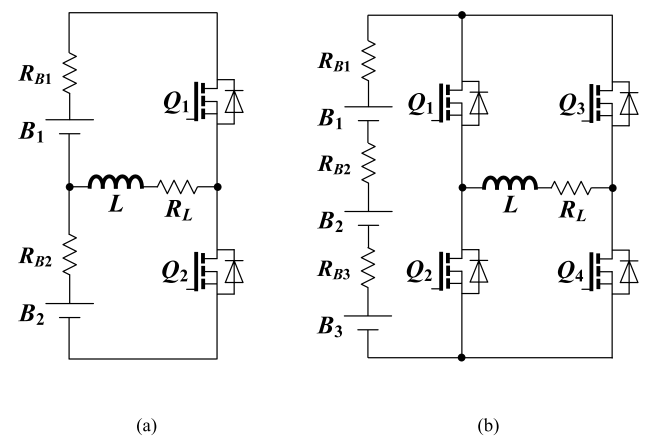


Fig. 1. The onboard AC heater based on buck-boost conversion. (a) The half-bridge-structured topology [30] for two series-connected cells. (b) The full-bridge-structured topology for three series-connected cells.

As shown in Fig. 2, taken the half-bridge- structured heater [30] as an example, the AC heater driven by two complementary PWM signals (i.e., PWM+ and PWM-) has four steady working states in one switching period. Along with the alternate operations of the four working states, a high-frequency AC current is automatically circulated between the two cells, based on which the batteries can be self-heated by the ohmic losses, i.e., $I^2_{B(RMS)}\cdot R_B$ and the electrochemical reaction. Fig. 2 (e) presents the theoretical waveforms of the AC heater. It can be clearly observed that in one switching period, States I and II realize respectively the charging and discharging for $B1$, and States III and IV achieve respectively the charging and discharging for $B2$. Moreover, during States II and III, energy is transferred from $B1$ to $B2$, and during States IV and I, energy is transferred back from $B2$ to $B1$. If the two cells have the same cell voltage and internal resistance, the transferred energy is equal, which ensures energy balancing between the two cells.

Based on Kirchhoff's current law (KCL), the AC current for cell B_1 during one switching period can be expressed as

$$\begin{cases} \frac{di_B}{dt} = -\frac{R_{eq}}{L}\cdot i_B + \frac{V_{OC}}{L}, & 0 < t < 0.5T_{SW} \\ i_B = 0, & 0.5T_{SW} < t < T_{SW} \end{cases}, \tag{1}$$

where L is the inductance of the inductor, V_{OC} is the Open-Circuit Voltage (OCV), T_{SW} is the switching period, and the equivalent resistance R_{eq} can be represented by

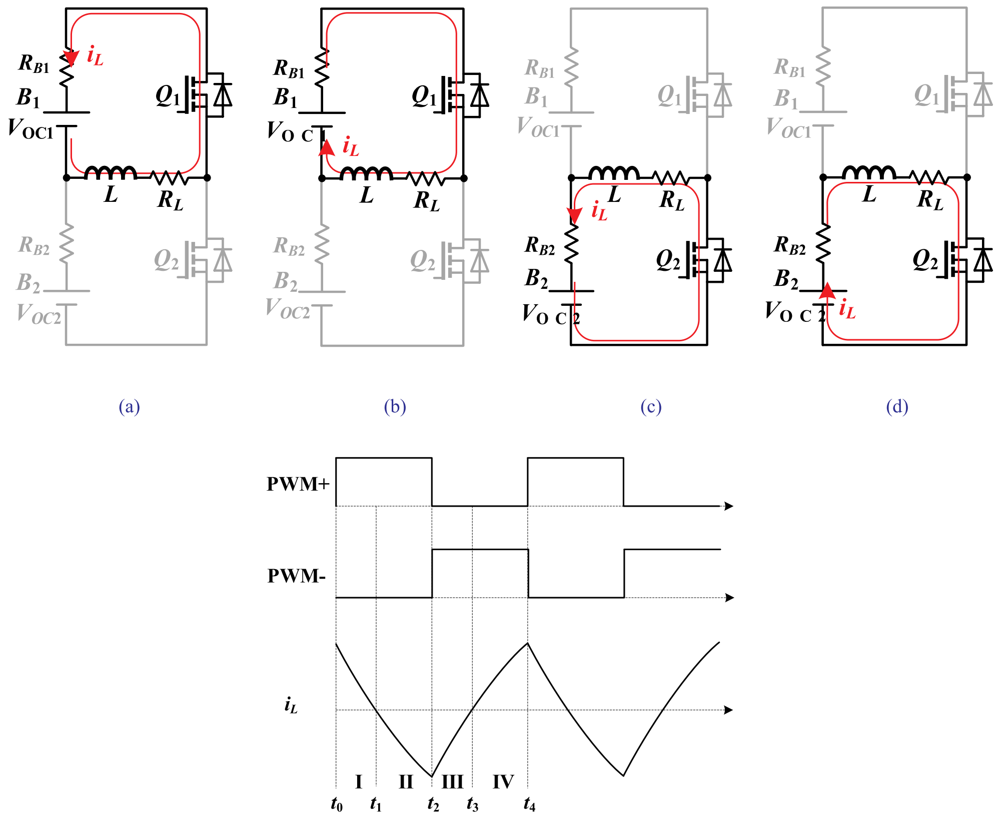
$$R_{eq} = R_L + R_{DS(on)} + R_B, \tag{2}$$

where R_{eq} is the equivalent resistance of the loop circuit (Ω), R_L is the resistance of the inductor (Ω), $R_{DS(on)}$ is the static drain-source on resistance of one MOSFET (Ω), and R_B is the internal resistance of one cell (Ω).

Assumed that the two cells have the same OCV and internal resistance, the amplitude of the charging current of B_1 is equal to that of the discharging current of B_1, that is,

$$i_B(0) = -i_B\left(\frac{T_{SW}}{2}\right). \tag{3}$$

According to (1) and (3), the AC current during one switching period can be derived:



(e)

Fig. 2. Operating states of the proposed AC heater for a two-cell battery pack [30]. (a) State I. (b) State II. (c) State III. (d) State IV. (e) Theoretical waveform.

$$\begin{cases} i_B(t) = \dfrac{V_{OC}}{R_{eq}} \cdot \left(1 - \dfrac{2}{1 + e^{-\frac{R_{eq}}{2L \cdot f_{SW}}}} e^{-\frac{R_{eq}}{L} t}\right), & 0 < t < \dfrac{T_{SW}}{2} \\ i_B(t) = 0, & \dfrac{T_{SW}}{2} < t < T_{SW} \end{cases}, \tag{4}$$

where f_{SW} is the switching frequency, which can be calculated by $f_{SW} = 1/T_{SW}$.

At $t = T_{SW}/2$, the maximum discharging current can be given by

$$I_{B(\max)} = i_B\left(\frac{T_{SW}}{2}\right) = \frac{V_{OC}}{R_{eq}} \cdot \left(\frac{2}{e^{-\frac{R_{eq}}{2L \cdot f_{SW}}} + 1} - 1\right). \tag{5}$$

Analogously, according to (4), the RMS battery heating current can be derived:

$$I_{B(RMS)} = \sqrt{f_{SW} \cdot \int_0^{T_{SW}} i_B^2(t) dt} = \frac{V_{OC}}{R_{eq}} \cdot \sqrt{\frac{1}{2} - \frac{2 \cdot L \cdot f_{SW}}{R_{eq}} \cdot \left(\frac{2}{1 + e^{-\frac{R_{eq}}{2L \cdot f_{SW}}}} - 1\right)}. \tag{6}$$

It can be observed that the maximum and RMS battery heating currents are in direct proportion to the OCV V_{OC}, inversely to the equivalent resistance R_{eq}, and can be increased by decreasing the switching frequency f_{SW} with the same inductance L or decreasing the inductance L at the same switching frequency f_{SW}.

It is worth mentioning that as shown in Fig. 2 (e), the batteries are discharged and charged once during one switching period. Therefore, the AC-heating frequency is equal to the switching frequency f_{SW}.

In summary, it can be noted that the high-frequency AC heater based on buck-boost conversion [30] can easily heat batteries at different AC-heating frequencies and RMS currents without the requirement of additional devices, which can be easily implemented in real EV applications.

2.2. Heating efficiency analysis

Because the proposed AC heater only uses battery energy to internally warm the batteries, the heating efficiency is a key parameter of the internal battery heating [30]. According to the analyses for the operating principle of the AC heater, the energy consumption during battery heating mainly includes two parts, i.e., the ohmic loss and the converter loss. As shown in Fig. 2, the ohmic resistance R_B and the inductor resistance R_L, and the on resistance $R_{DS(on)}$ are in a current loop. Thus, the heating efficiency for one cell can be given by Ref. [30].

$$\eta_h = \frac{R_B \cdot I_{L(\mathrm{RMS})}^2}{(R_B + R_L + R_{DS(on)}) \cdot I_{L(\mathrm{RMS})}^2} \times 100\% = \frac{R_B}{R_B + R_L + R_{DS(on)}} \times 100\%. \tag{7}$$

where $I_{L(RMS)}$ is the RMS current of the inductor. It can be seen that the larger the ohmic resistance R_B, or the smaller the inductor resistance R_L and the on resistance $R_{DS(on)}$, the higher the heating efficiency. In fact, the battery ohmic resistance R_B mainly depends on the temperature and switching frequency. It is mentioning that the switching frequency can be optimized to achieve a larger ohmic resistance, which will be detailedly discussed below. In addition, high-performance inductors and MOSFETs with small static drain-source on resistance should be chosen to achieve a smaller equivalent resistance R_1.

2.3. Battery one-order equivalent circuit model

In order to maximize the battery ohmic resistance to improve the heating speed and efficiency, the common one-order equivalent circuit model [35] is introduced and shown in Fig. 3. It can be seen that the one-order equivalent circuit model mainly includes a Resistor-Capacitor (R_1C_1) branch, an ohmic resistor R_B, an inductor L_b. and a voltage source V_{OC}, which are connected in series. R_1 presents the electrochemical polarization resistance. C_1 presents the electric double-layer capacitance. L_b presents the battery inductance.

According to Kirchhoff's voltage law (KVL), at lower switching frequencies, the battery impedance can be given by Ref. [35].

$$Z(T,f_H)=\left[R_B(T)+\frac{R_1(T)}{1+(2\pi f_{SW})^2R_1^2(T)C_1^2}\right]-j\frac{2\pi f_{SW}R_1^2(T)C_1}{1+(2\pi f_{SW})^2R_1^2(T)C_1^2}, \tag{8}$$

where $R_B(T)$ and $R_1(T)$ are functions of the battery temperature T.

At higher switching frequencies, the effect of the battery inductance L_b cannot be neglected, the battery impedance can be derived as [35].

$$\begin{aligned}Z(T,f_{SW})=&\left[R_B(T)+\frac{R_1(T)}{1+(2\pi f_{SW})^2R_1^2(T)C_1^2}\right]\\&+j\left[2\pi f_{SW}L_b-\frac{2\pi f_{SW}R_1^2(T)C_1}{1+(2\pi f_{SW})^2R_1^2(T)C_1^2}\right].\end{aligned} \tag{9}$$

From (8) and (9), it can be noted that the battery temperature T and switching frequency f_{SW} are the two major factors determining the battery impedance. Not only at lower switching frequencies, the battery impedance will increase with the decrease of the switching frequency, but also at higher switching frequencies, the battery impedance will also increase with the increase of the switching frequency because of the effect of battery inductance L_b.

2.4. Thermoelectric model of high-frequency AC heating

In order to estimate the heating speed at different AC-heating frequencies and RMS currents, a high-frequency heating model considering the heat generation of the charge transport is developed, which also provides a guidance for the design of the buck-boost-based AC heater [30].

The proposed high-frequency model consists of an electrochemical-thermal coupling module for the battery AC charging/discharging and a thermal module for the high-frequency lithium ion transport, which can be expressed as

$$m\cdot c\cdot\frac{dT}{dt}+h\cdot S\cdot(T-T_0)=I^2_{B(RMS)}\cdot R_B+k\cdot f_{SW}\cdot I_{B(RMS)}, \tag{10}$$

where m is the battery mass, c is the specific heat capacity, S is the surface area of the battery, h is the equivalent heat transfer coefficient, T_0 is the ambient temperature, and $h\cdot S\cdot(T-T_0)$ represents the heat diffusion to the environment. It can be observed that the larger the temperature difference, the more the heat diffusion to the environment. $I^2_{B(RMS)}\cdot R_B$ represents the heat generation of the ohmic resistance. Because the heat generation of the lithium ion transport mainly depends on the heating frequency and current, the term $k\cdot f_{SW}\cdot I_{B(RMS)}$ can be used to reasonably represent the heat generation, and here k is the high-frequency heating coefficient. It is important to note that the heat generation of the charge transport could be neglected when the AC-heating frequency is very low. Therefore, (10) can be simplified as the temperature-rise model proposed in Ref. [18] at a low switching frequency. This means that the proposed thermoelectric model can simulate the temperature rise of the all-frequency AC heating.

According to (10), the high-frequency heating coefficient can be expressed as

$$k=\frac{m\cdot c\cdot\frac{dT}{dt}+h\cdot S\cdot(T-T_0)-I^2_{B(RMS)}\cdot R_B}{f_{SW}\cdot I_{B(RMS)}}, \tag{11}$$

where dT/dt is the battery temperature-rise rate, which can be directly measured at the battery temperature T.

Further, combining (6) and (10), the temperature-rise model of the high-frequency AC heating can be represented by

$$\begin{aligned}m\cdot c\cdot\frac{dT}{dt}+h\cdot S\cdot(T-T_0)=V_{OC}^2\cdot\frac{R_B}{R_{eq}^2}\cdot\left\{\frac{1}{2}-\frac{2\cdot L\cdot f_{SW}}{R_{eq}}\cdot\left(\frac{2}{1+e^{-\frac{R_{eq}}{2L\cdot f_{SW}}}}-1\right)\right\}\\+k\cdot f_{SW}\cdot I_{B(RMS)}.\end{aligned} \tag{12}$$

By solving (12), T can be expressed as

$$T=\left(T_0-\frac{h\cdot S\cdot T_0+A}{h\cdot S}\right)\cdot e^{-\frac{h\cdot S}{m\cdot c}t}+\frac{h\cdot S\cdot T_0+A}{h\cdot S}, \tag{13}$$

where

$$A=V_{OC}^2\cdot\frac{R_B}{R_{eq}^2}\cdot\left\{\frac{1}{2}-\frac{2\cdot L\cdot f_{SW}}{R_{eq}}\cdot\left(\frac{2}{1+e^{-\frac{R_{eq}}{2L\cdot f_{SW}}}}-1\right)\right\}+k\cdot f_{SW}\cdot I_{B(RMS)}. \tag{14}$$

According to the above analysis, it can be observed that with the same inductance L, the RMS heating current can be increased by decreasing the switching frequency, leading to more heat generation of the ohmic loss. On the contrary, increasing the AC-heating frequency can significantly increase the heat generation of the ohmic loss and charge transport.

It is important to note that a too large RMS heating current may cause lithium ion deposition, reducing battery available capacity. Therefore, a higher AC-heating frequency is recommended to achieve a good trade-off between the heating speed and negative effect on battery health.

3. Results and discussion

3.1. Experiment setup

A prototype for two $LiNiMnCoO_2$ battery cells was built, which included a controller dSPACE, a computer, a monitoring interface, a

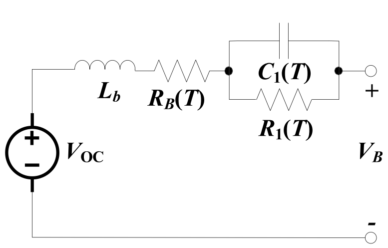


Fig. 3. The common one-order equivalent circuit model [35].

Table 1
Parameters of proposed heating system [30].

Parameter	Symbol	Value
Battery type	–	Cylindrical 18650
Nominal voltage	V	3.7 V
Battery capacity	C	2500 mAh
Battery mass	m	46.0 g
Battery surface area	S	4.18×10^{-3} m^2
Specific heat capacity	c	1.72 J g^{-1} k^{-1}
Equivalent heat transfer coefficient	h	21.6 W m^{-2} k^{-1}
MOSFET type	Q	STP220N6F7
Drain-source on resistance of MOSFETs	$R_{DS(on)}$	2.4 mΩ
Inductance	L	2.1–98.5 μH
Resistance of the inductor	R_L	10–100 mΩ

temperature chamber, an oscilloscope, the proposed AC heating converter, $LiNiMnCoO_2$ batteries, and so on. Table 1 detailedly presents the parameters of the proposed heating system [30]. It is worth mentioning that before heating, the $LiNiMnCoO_2$ batteries were precooled for more than 3 h by the temperature chamber, whose ambient temperature was set as −20 °C. When the cold batteries were heated to 0 °C or the heating time was beyond an hour, the AC heater was disabled.

3.2. Battery internal resistances and analysis

Fig. 4 presents the internal resistance of a $LiNiMnCoO_2$ battery versus the AC frequency at different ambient temperatures. It can be noted that the resistance curve is approximatively an upward parabola with larger values at both the ends and smaller values in the middle. The minimum internal resistance of the $LiNiMnCoO_2$ battery is achieved at about 10 kHz, which will sharply increase with the decrease or increase of the AC frequency from 10 kHz. Particularly at lower ambient temperatures, the increase rate of the internal resistance significantly increases. In summary, not only decreasing the heating frequency but also increasing the AC heating frequency can significantly increase the heat generation of the ohmic resistance. The variation trends of the internal resistance versus the AC frequency at different ambient temperatures are consistent with (9).

3.3. Experimental waveforms and analysis

Fig. 5 shows the measured waveforms of the high-frequency AC heater at the switching frequency of 10 kHz with $L = 9$ μH. We can observe that the AC current is approximatively a triangle waveform. During the first half of the switching period, the heating current decreases approximately linearly from 10.86 A to −10.32 A, which achieves the charging and discharging for B_1. During the second half of the switching period, the heating current increases approximately linearly from −10.32 A to 10.86 A, which realizes the charging and discharging for B_2. Therefore, each cell can be heated once during one switching period. It is worth mentioning that the RMS heating current $I_{B(RMS)}$ for one cell at 10 kHz is about 4.17 A, i.e., 1.7 C. In addition, the calculated AC current (i.e., the red dotted line) agrees well with the measured current (i.e., the blue full line), which proves the validity of (4).

3.4. Heating results and analysis at different switching frequencies

Fig. 6 presents the AC heating results at different switching frequencies with almost the same RMS heating current $I_{B(RMS)}$, i.e., 3.81 A (1.52 C) under the ambient temperature of −20 °C. Table 2 summarizes

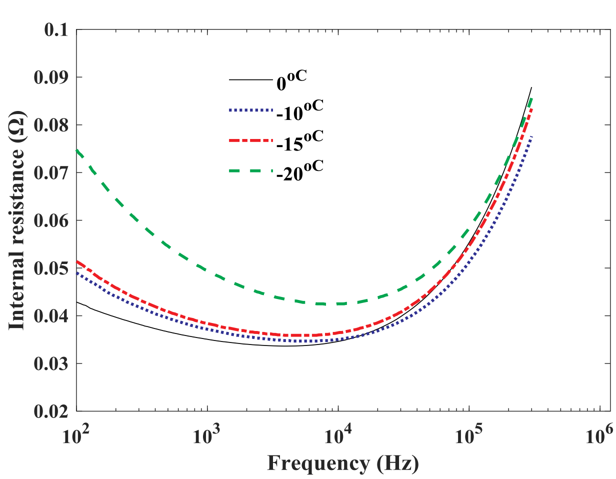


Fig. 4. The internal resistance of a $LiNiMnCoO_2$ battery versus the AC frequency at different ambient temperatures.

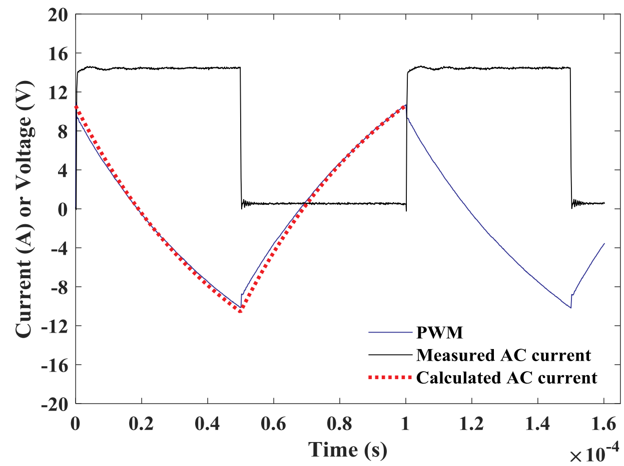


Fig. 5. Measured and calculated waveforms of the AC heater at $f_{SW} = 10$ kHz.

the comparison of the heating parameters and results. It can be observed that the heating time is dramatically reduced by increasing the AC-heating frequency due to the more heat generation of the charge transport. Particularly, when the AC-heating frequency is set as $f_{SW} = 0.83$ kHz, the AC heater needs a 13.2-min heating time and 7.3% energy loss to warm batteries from −20 °C to 0 °C, whose average rising rate of temperature is only 1.52 °C/min. By comparison, when the AC-heating frequency increases to 45 kHz, the heating time is shortened to 7.4 min and the rising rate of temperature reaches 2.70 °C/min. Moreover, the energy loss is reduced to about 6.0% because of the least heat diffusion to the environment.

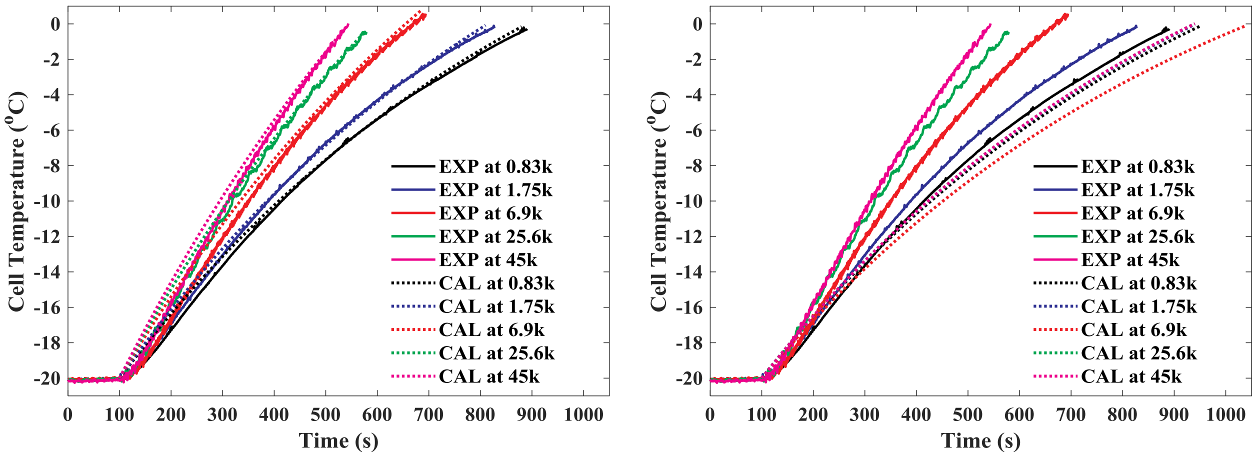
As shown in Fig. 6(a), the thermoelectric model considering the heat generation of the charge transport can fit well the measured values at different AC-heating frequencies. As shown in Table 2, the predicted heating times of the proposed thermoelectric model at different AC-heating frequencies are also consistent with the measured heating times. This proves the validity of (13).

However, as shown in Fig. 6(b), the conventional heating model without the heat generation of the high-frequency charge transport misses the measured values, especially, at higher AC-heating frequencies. This is also verified by that the conventional model has an increasing predicted error in terms of the heating time with the increase of the AC-heating frequency shown in Table 2. It is worth mentioning that the conventional model can accurately predict the experimental values at the lower AC-heating frequency, i.e., 0.83 kHz due to the less heat generation of the charge transport. This is consistent with the results in Ref. [18].

It is important to note that as shown in Table 2, the high-frequency heating coefficient k decreases with the increase of the AC-heating frequency. This means that there is theoretically an optimal AC-heating frequency to obtain the maximum heat generation of the charge transport.

3.5. Heating results and analysis with different RMS heating currents

Fig. 7 shows the heating results with different RMS heating currents at the AC-heating frequency of $f_{SW} = 10$ kHz under the ambient temperature of −20 °C. Table 3 summarizes the comparison of the heating parameters and results. It can be observed that the heating time is significantly reduced by increasing the RMS current. When the RMS heating current $I_{B(RMS)}$ is 1.2 C, the heating time is up to 22.5 min with the average temperature-rise rate of 0.89 °C/min. This heating consumes higher battery energy, i.e., about 10%. When the RMS current $I_{B(RMS)}$ increases to 2.0 C, the heater only requires a 5.6-min heating time with the average temperature-rise rate of 3.57 °C/min and 5.7% energy consumption, much more efficient than the lower RMS current.

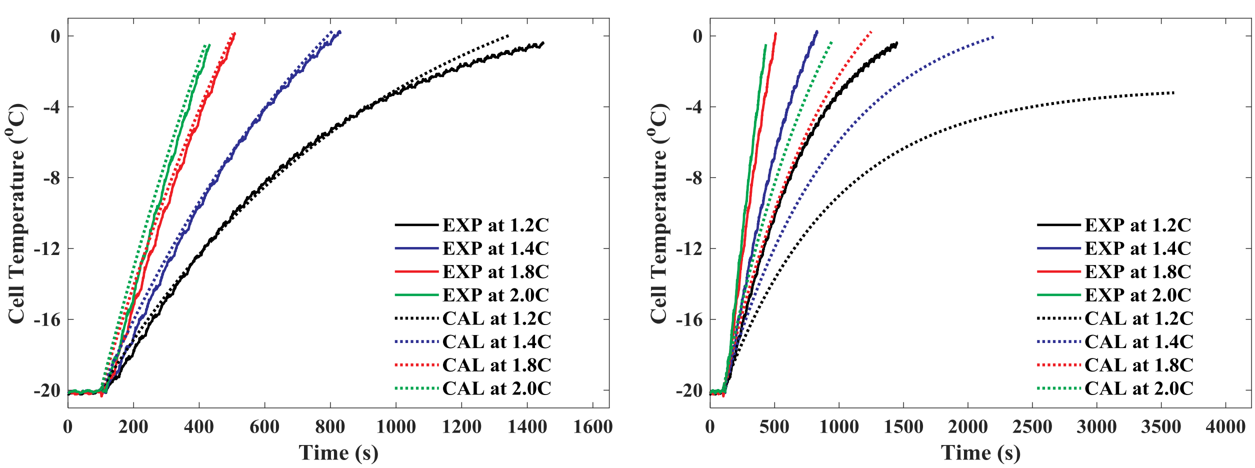


(a)   (b)

Fig. 6. The comparative AC heating results for two $LiNiMnCoO_2$ cells at different AC-heating frequencies with the same RMS heating current under the ambient temperature of $-20\,^{\circ}C$. (a) The proposed thermoelectric model considering the heat generation of the charge transport. (b) The conventional heating model without the heat generation of the charge transport.

Table 2
A comparison of the AC heating at different AC-Heating frequencies with the same RMS heating current of 1.52 C.

AC-heating frequency (kHz)	0.83	1.75	6.9	25.6	45
Inductance (μH)	98.5	50.1	13.1	3.7	2.1
Temperature rise rate (°C/min)	1.52	1.65	2.02	2.50	2.70
Measured heating time (min)	13.2	12.1	9.9	8	7.4
Predicted heating time of the conventional heating model (min)	14.2	14	15.7	14	14
Predicted heating time of the proposed heating model (min)	13	11.8	9.8	7.9	7.4
Heating coefficient k ($J.kHz^{-1}.A^{-1}.s^{-1}$)	4.74×10^{-5}	4.7410^{-5}	4.26×10^{-5}	1.32×10^{-5}	0.94×10^{-5}
Energy loss (%)	7.3	7.1	7.0	6.4	6.0



(a)   (b)

Fig. 7. The comparative internal-heating results for two $LiNiMnCoO_2$ cells with different RMS heating currents at the AC-heating frequency of 10 kHz. (a) With the heat generation of battery electrochemical reaction. (b) Without the heat generation of the charge transport.

However, a larger RMS heating current may reduce battery lifetime.

Fig. 7 (a) and (b) present a comparison of the temperature fittings with and without the heat generation of the high-frequency charge transport at different RMS heating currents. It can be observed that the proposed thermoelectric model considering the heat generation of the charge transport can fit well the measured values at different RMS heating currents, and the predicted heating times are close to the measured values. This proves the validity of (13). It can be noted that the heating coefficient k shown in Table 3 increases with the increase of the RMS heating current $I_{B(RMS)}$ at the same AC-heating frequency.

However, as shown in Fig. 7(b), the conventional heating model without the heat generation of the charge transport cannot accurately predict the temperature and heating time. The predicted heating time dramatically mismatches the measured values. Nevertheless, the prediction error decreases with the increase of the RMS current because of the increased heat generation of the ohmic resistance. This shows that

Table 3
A Comparison of the AC Heating with Different RMS heating currents at the AC-Heating Frequency of 10 kHz.

RMS heating current (C)	1.2	1.4	1.8	2.0
Inductance (μH)	14.2	12.1	9.8	8.6
Temperature rise rate (°C/min)	0.89	1.64	2.94	3.57
Measured heating time (min)	22.5	12.2	6.8	5.6
Predicted heating time of the conventional heating model (min)	More than 60	35	19.2	14.2
Predicted heating time of the proposed heating model (min)	20.8	11.8	6.7	5.3
Heating coefficient k (J. $kHz^{-1}.A^{-1}.s^{-1}$)	2.8×10^{-5}	3.8×10^{-5}	5.4×10^{-5}	5.8×10^{-5}
Energy loss (%)	10	7.6	6.2	5.7

the conventional heating model has poor prediction accuracy with different RMS heating currents at the AC-heating frequency of 10 kHz and above.

3.6. Heating speed analysis

According to the above analysis, Fig. 8 (a) and (b) further present a comparison of the heating times versus the AC-heating frequency and RMS heating current, respectively. It can be observed that the heating speed dramatically increases with the increase of the AC-heating frequency or the RMS heating current. However, the increment of the heating speed decreases with the increase of the AC-heating frequency or the RMS heating current. By comparison, it is important to note that increasing the RMS heating current is more effective for improving the heating speed than increasing the AC-heating frequency. However, if the RMS current is too large, it may cause lithium ion deposition, reducing battery life. As a consequence, the AC-heating frequency should be as high as possible to achieve a fast heating speed without detrimental effect on battery health.

3.7. Comparison with conventional heating approaches

In order to analyze the strengths and weaknesses of the existing heating solutions, this section presents a comparison of the proposed high-frequency heating strategy with three conventional heating approaches in terms of power supply, efficiency, speed, reliability, potential harm to battery, heating consistency, and implementation. The comprehensive comparison is shown in Table 4.

The external heating approach [13] needs electric resistance wires and external power source to heat batteries from outside to inside. Therefore, low heat conduction efficiency, long heating time, bad temperature consistency, and difficult implementation in EVs are the prominent weaknesses of this solution.

The low-frequency AC heating strategy [18] uses an external device, i.e., Autolab PGSTAT302 N, to generate a low-frequency sinusoidal current to internally heat batteries based on the heat generation of the ohmic loss. The lower AC heating frequency leads to a slow heating speed, much harm to batteries, large size, high cost, and difficult implementation in vehicles.

Other than the traditional lithium-ion batteries, the self-heating battery [26] has an activation terminal, which is connected with a metal foil inserted in the battery. When the activation and negative terminals are short circuited, a mass of heat is generated inside the battery and is fastly transferred between the electrodes and electrolyte. Obviously, the primary advantages of the self-heating battery [26] are fast heating speed, high efficiency, and the small size and low cost because of the nonuse of additional heating devices. However, this new battery still needs to be further tested and verified in terms of the reliability and safety, which can not be directly implemented in EVs.

Compared with the low-frequency AC heating approach [18], the proposed heating strategy generates a high-frequency AC current to internally warm cold batteries. A lot of heat is generated by both the ohmic loss and charge transport, leading to a fast heating speed and high efficiency. Moreover, due to the high switching frequency, the size and cost of the AC heater can be significantly reduced, resulting in a simple implementation in vehicles. It is worth mentioning that the high-frequency AC heating effectively restrains lithium ion separation and produces little harm to batteries.

4. Conclusion

Due to the high heating speed, efficiency, and good consistency, the AC heating is very promising to improve the performance of lithium-ion batteries at low temperatures. However, the existing AC heating approach works at low switching frequencies, leading to a bulky size and high cost, which is difficult to be applied to vehicles. Therefore, this paper develops a high-frequency AC heating strategy based on buck-boost conversion and the corresponding thermoelectric model considering the heat generation of the charge transport at high AC-heating frequencies. A prototype for two $LiNiMnCoO_2$ cells was built. The proposed high-frequency AC heater and thermoelectric model were

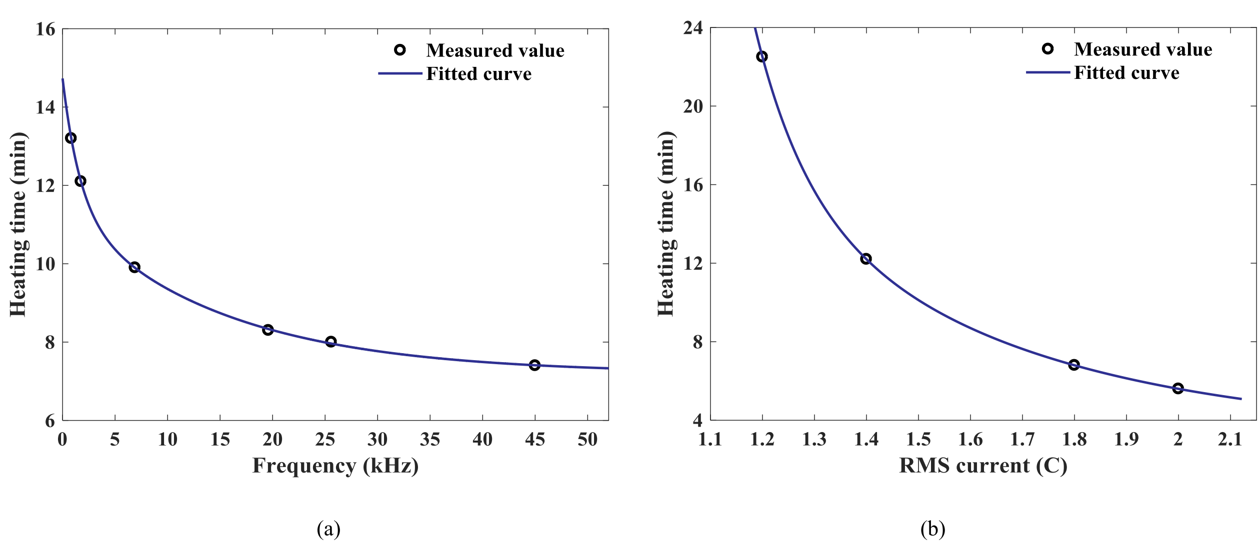


Fig. 8. (a) The heating time versus the AC-heating frequency. (b) The heating time versus the RMS heating current.

Table 4
A comparison of conventional heating solutions.

Heating approaches	Power supply	Efficiency	Speed	Reliability	Harm to battery	Consistency	Implementation
External heating [13]	External device	Low	Slow	Low	Much	Bad	Complex
Low-frequency AC heating [18]	External device	Low	Slow	Low	Much	Good	Complex
Self-heating battery [26]	Battery	High	Fast	High	Little	Good	Complex
Proposed high-frequency strategy	Battery	High	Fast	High	Little	Good	Simple

validated at different AC-heating frequencies and RMS heating currents. Through detailed analyses of experimental results, some conclusions are obtained as follows. At higher AC-heating frequencies, the heat generation of the lithium ion transport cannot be neglected in battery internal heating. Particularly, increasing the AC-heating frequency at the same RMS heating current can dramatically improve the heating speed and efficiency without causing further damage to batteries. In order to improve the heating speed, increasing the heating current is more effective than increasing the AC-heating frequency, nevertheless which may cause lithium ion plating and reduce battery life. In summary, this study indicated that the AC-heating frequency and current should be seriously considered to achieve a fast heating speed with less negative effect on battery performance. In the future, the optimal AC-heating frequency and current will be further studied to achieve a good trade-off between the heating speed and harm to batteries.

Declaration of competing interest

The authors declare no conflict of interest.

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References

[1] W. Zhou, L. Yang, Y. Cai, T. Ying, Dynamic programming for new energy Vehicles based on their work modes part I: electric vehicles and hybrid electric vehicles, J. Power Sources 406 (2018) 151–166.
[2] J. Sieg, J. Bandlow, T. Mitsch, D. Dragicevic, T. Materna, B. Spier, H. Witzenhausen, M. Ecker, D.U. Sauer, Fast charging of an electric vehicle lithium-ion battery at the limit of the lithium deposition process, J. Power Sources 427 (2019) 260–270.
[3] F. Feng, R. Lu, S. Zhang, C. Zhu, G. Wei, Temperature Characteristics Research on Lifepo$_4$ Cells Series Battery Pack In Electric Vehicles, Springer Berlin Heidelberg, 2014.
[4] J. Yi, U.S. Kim, C.B. Shin, T. Han, S. Park, Modeling the temperature dependence of the discharge behavior of a lithium-ion battery in low environmental temperature, J. Power Sources 244 (2013) 143–148.
[5] G. Zhang, S. Ge, X. Yang, Y. Leng, D. Marple, C. Wang, Rapid restoration of electric vehicle battery performance while driving at cold temperatures, J. Power Sources 371 (2017) 35–40.
[6] Extreme temperatures affect electric vehicle driving range, AAA says. http://newsroom.aaa.com/2014/03/extreme-temperatures- affect-electric-vehicle-driving-range-aaa-says/, Mar. 2014. Online. Accessed on 03/02/2018.
[7] T. Waldmann, M. Wilka, M. Kasper, M. Fleischhammer, M. Wohlfahrt-Mehrens, Temperature dependent ageing mechanisms in lithium-ion batteries-a post-mortem study, J. Power Sources 262 (2014) 129–135.
[8] S. Herreyre, O. Huchet, S. Barusseau, F. Perton, J.M. Bodet, Ph Biensan, New Li-ion electrolytes for low temperature applications, J. Power Sources 97 (2001) 576–580.
[9] G. Zhu, K. Wen, W. Lv, X. Zhou, Y. Liang, F. Yang, Z. Chen, M. Zou, J. Li, Y. Zhang, W. He, Materials insights into low-temperature performances of lithium-ion batteries, J. Power Sources 300 (2015) 29–40.
[10] P. Wu, J. Romberg, H. Ge, Y. Zhang, J. Zhang, Electrochemical-Thermal Coupled Model of Lithium-Ion Batteries for Low Temperature Charging, Springer Singapore, 2017.
[11] T. Wang, X. Wu, S. Xu, H. Hofmann, J. Du, J. Li, M. Ouyang, Z. Song, Performance of plug-in hybrid electric vehicle under low temperature condition and economy analysis of battery pre-heating, J. Power Sources 401 (2018) 245–254.
[12] A.-A. Pesaran, A. Vlahinos, T. Stuart, Evaluation of Heating Methods for HEV Batteries for Cold Seasons, NREL Milestone Report, National Renewable Energy Laboratory, 2001.
[13] A. Vlahinos and A. A. Pesaran, Energy Efficient Battery Heating in Cold Climates, Society of Automotive Engineers, pp. 1–8, 2002-01-1975.
[14] Y. Yang, X. Wei, Y. Liu, H. Dai, J. Zhu, Q. Fang, A research on the AC heating of automotive lithium-ion battery, Automot. Eng. 38 (7) (Jul. 2016) 901–908.
[15] A.-A. Pesaran, A. Vlahinos, T. Stuart, Cooling and Preheating of Batteries in Hybrid Electric Vehicles, in 6th ASME-JSME Thermal Engineering Joint Conference, 2003, pp. 1–7.
[16] Y. Ji, C.Y. Wang, Heating strategies for Li-ion batteries operated from subzero temperatures, Electrochim. Acta 107 (2013) 664–674.
[17] X. Yang, T. Liu, C. Wang, Innovative heating of large-size automotive Li-ion cells, J. Power Sources 342 (2017) 598–604.
[18] J. Zhang, H. Ge, Z. Li, Z. Ding, Internal heating of lithium-ion batteries using alternating current based on the heat generation model in frequency domain, J. Power Sources 273 (2015) 1030–1037.
[19] J. Jiang, H. Ruan, B. Sun, W. Zhang, W. Gao, L. Wang, L. Zhang, A reduced low-temperature electro-thermal coupled model for lithium-ion batteries, Appl. Energy 177 (2016) 804–816.
[20] S. Mohan, Y. Kim, A.G. Stefanopoulou, Energy-conscious warm-up of li-ion cells from subzero temperatures, IEEE Trans. Ind. Electron 63 (2016) 2954–2964.
[21] H. Ruan, J. Jiang, B. Sun, W. Zhang, W. Gao, L.Y. Wang, Z. Ma, A rapid low-temperature internal heating strategy with optimal frequency based on constant polarization voltage for lithium-ion batteries, Appl. Energy 177 (2016) 771–782.
[22] J. Zhu, Z. Sun, X. Wei, H. Dai, W. Gu, Experimental investigations of an AC pulse heating method for vehicular high power lithium-ion batteries at subzero temperatures, J. Power Sources 367 (2017) 145–157.
[23] G. Zhang, S. Ge, X. Yang, Y. Leng, D. Marple, C. Wang, Rapid restoration of electric vehicle battery performance while driving at cold temperatures, J. Power Sources 371 (2017) 35–40.
[24] J. Jiang, H. Ruan, N. Cui, B. Sun, L. Wang, W. Gao, W. Zhang, A low-temperature internal heating strategy without lifetime reduction for large-size automotive lithium-ion battery pack, Appl. Energy 230 (2018) 257–266.
[25] S. Guo, R. Xiong, K. Wang, F. Sun, A novel echelon internal heating strategy of cold batteries for all-climate electric vehicles application, Appl. Energy 219 (2018) 256–263.
[26] C. Wang, G. Zhang, S. Ge, T. Xu, Y. Ji, X. Yang, Y. Leng, Lithium-ion battery structure that self-heats at low temperatures, Nature 529 (2016) 515–518.
[27] G. Zhang, S. Ge, T. Xu, X.-G. Yang, H. Tian, C.-Y. Wang, Rapid self-heating and internal temperature sensing of lithium-ion batteries at low temperatures, Electrochim. Acta 218 (2016) 149–155.
[28] T.A. Stuart, A. Hande, HEV battery heating using AC currents, J. Power Sources 129 (2004) 368–378.
[29] Y. Shang, K. Liu, N. Cui, Q. Zhang, C. Zhang, A sine-wave heating circuit for automotive battery self-heating at subzero temperatures, IEEE Trans. Ind. Informatics (2019).
[30] Y. Shang, B. Xia, N. Cui, C. Zhang, C. Mi, An automotive onboard AC heater without external power supplies for lithium-ion batteries at low temperatures, IEEE Trans. Power Electron. 33 (2018) 7759–7769.
[31] M. Xiao, S.-Y. Choe, Theoretical and experimental analysis of heat generations of a pouch type $LiMn_2O_4$/carbon high power Li-polymer battery, J. Power Sources 241 (2013) 46–55.
[32] G. Liu, M. Ouyang, L. Lu, J. Li, X. Han, Analysis of the heat generation of lithium-ion battery during charging and discharging considering different influencing factors, J. Therm. Anal. Calorim. 116 (2014) 1001–1010.
[33] T. Dong, P. Peng, F. Jiang, Numerical modeling and analysis of the thermal behavior of NCM lithium-ion batteries subjected to very high C-rate discharge/charge operations, Int. J. Heat Mass Transf. 117 (2018) 261–272.
[34] Y. Yang, X. Wei, Y. Liu, H. Dai, J. Zhu, Q. Fang, A research on the AC heating of automotive lithium-ion battery, Automot. Eng. 38 (2016) 901–908.
[35] Y. Shang, C. Zhu, Y. Fu, C. Mi, An integrated heater equalizer for lithium-ion batteries of electric vehicles, IEEE Trans. Ind. Electron 66 (2018) 4398–4405.